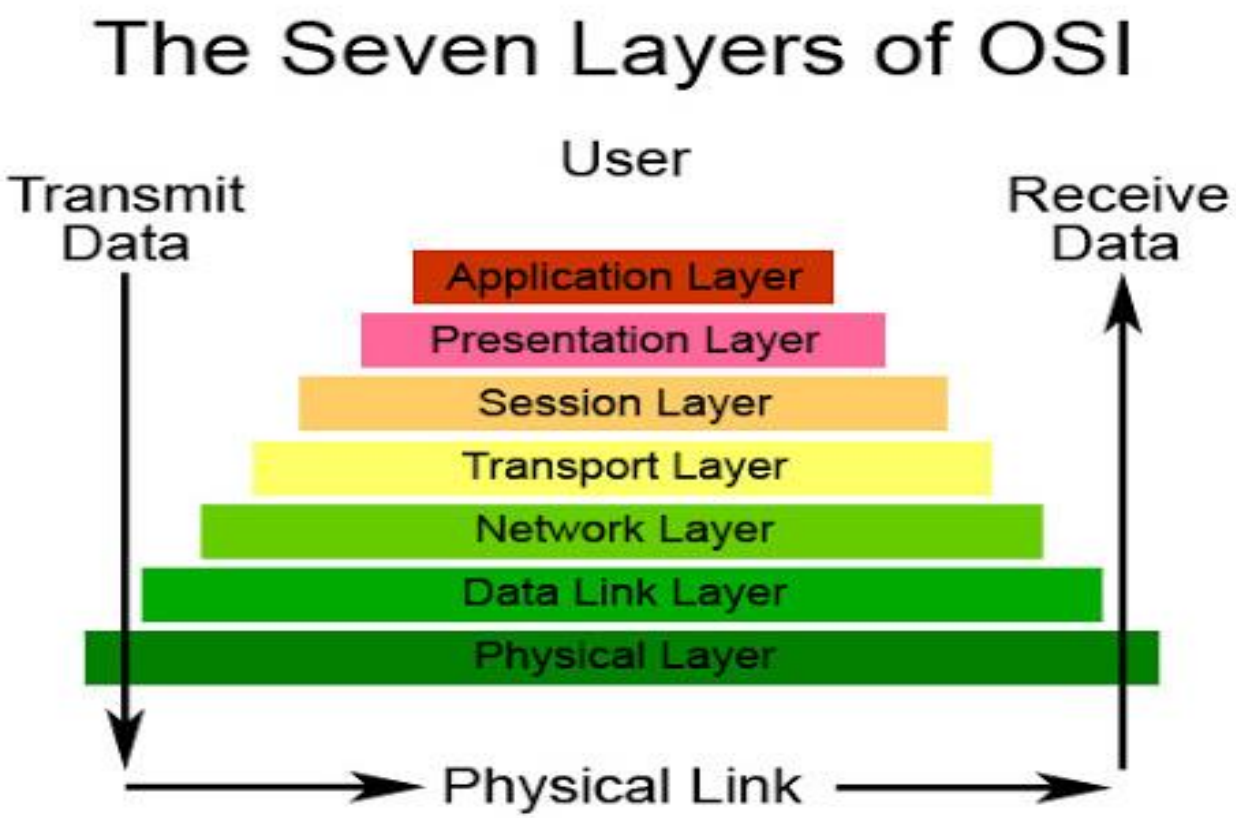
WEEK 1: Networking Fundamentals | Days: 1-7

Day 1 Tasks—OSI MODEL

➢ Draw the complete 7-layer OSI model diagram?

The OSI (Open Systems Interconnection) model is a conceptual framework developed by the International Organization for Standardization (ISO). It provides a structured approach to understanding how data travels from one device to another over a network.

Layer	Name Function	Example
7	Application	User interface HTTP, FTP, DNS
6	Presentation	Data translation, encryption SSL/TLS, JPEG
5	Session	Manage sessions NetBIOS, RPC
4	Transport	Reliable/unreliable delivery TCP, UDP
3	Network	Routing, logical addressing IP, and Router
2	Data Link	Physical addressing, error detection, Switch
1	Physical	Bits, signals, cables RJ45, Fiber



Layer 1: Physical Layer

This is the bottommost layer of the OSI reference model. It handles the physical connection between communicating devices.

- Responsible for transmitting raw bits from one node to the next.
- Upon receiving data, it converts incoming signals into binary form (0s and 1s) and forwards them to the Data Link layer.
- Typical devices operating at this layer include hubs, repeaters, modems, and various types of cabling.

Layer 2: Data Link Layer (DLL)

The Data Link Layer handles node-to-node message delivery. It ensures error-free data transfer between two directly connected nodes over the physical layer.

- When a packet reaches a network, the DLL is responsible for forwarding it to the correct host using the device's MAC address.
- At this layer, a packet is called a frame.
- Common DLL devices include switches and bridges.
- The packet received from the Network Layer is broken into smaller frames based on the frame size supported by the Network Interface Card (NIC).
- The DLL adds the sender and receiver MAC addresses into the frame header.

Sublayers of the Data Link Layer

- **Logical Link Control (LLC):** Coordinates communication between upper layers and handles error checking and flow control.-
- **Media Access Control (MAC):** Regulates how devices access the shared transmission medium and manages MAC addressing.

Layer 3: Network Layer

The Network Layer handles data transmission between hosts located in different networks. It is also responsible for routing packets across the network.

- The sender and receiver IP addresses are inserted into the header by the Network Layer.
- At this layer, a segment is referred to as a packet.
- The Network Layer is implemented using networking devices such as routers and layer-3 switches.

Handles logical addressing and routing of packets.

- Uses IP addresses for routing decisions
- Determines the best path to the destination
- Manages packet forwarding and routing tables

Layer 4: Transport Layer

The Transport Layer ensures end-to-end message delivery. It provides services to the Application Layer and receives services from the Network Layer.

- Data at this layer is called a segment.
- The Transport Layer guarantees successful data transmission.
- Common Transport Layer protocols include TCP, UDP,NetBIOS and PPTP
- It adds source and destination port numbers to the segment header and forwards the segmented data to the Network Layer.

Ensures reliable or fast delivery of data.

- Uses TCP (reliable) and UDP (fast, connectionless)
- Handles segmentation and reassembly of data
- Provides flow control (sliding window)
- Ensures error recovery and end-to-end communication

Layer 5: Session Layer

The Session Layer manages the establishment, maintenance, and termination of communication sessions between two devices. It also provides authentication and security features.

- Establishes, manages, and terminates sessions.
- Controls dialog between devices (duplex/half-duplex).
- Maintains session checkpoints and recovery.
- Handles session authentication and reconnection.
- Used in RPC, SQL sessions, NetBIOS.

Layer 6: Presentation Layer

The Presentation Layer is also known as the Translation Layer. It extracts data from the Application Layer and transforms it into the appropriate format for network transmission.

- Common Presentation Layer protocols include TLS/SSL (Transport Layer Security / Secure Sockets Layer).
- Formats such as JPEG, MPEG, and GIF are encoding standards managed by this layer.

Translates, encrypts, and formats data.

- Converts data formats (encoding/decoding).
- Applies encryption and decryption (SSL/TLS).
- Handles compression for efficient transmission.
- Makes data readable for the application layer.

Layer 7: Application Layer

The Application Layer sits at the top of the OSI model stack. It is implemented by network applications that generate the data to be sent across the network.

- This layer acts as a gateway for application services to access the network and display information.
- Supports protocols like HTTP, FTP, SMTP, and DNS.
- Provides network services such as email, web access, and file transfer.
- Interacts directly with user applications.
- Identifies communication partners and resource availability

Data Flow in the OSI Model

When information is transferred from one device to another, it passes through all seven layers of the OSI model. First, data travels downward through the seven layers at the sender's side, then climbs back up the seven layers at the receiver's side.

Step-by-Step Data Flow

Layer	Action
Application Layer	Applications generate the data.
Presentation Layer	Data is formatted and encrypted.
Session Layer	Connections are established and managed.
Transport Layer	Data is split into segments for reliable delivery.
Network Layer	Segments are packaged into packets and routed.
Data Link Layer	Packets are framed and sent to the next device.
Physical Layer	Frames are converted into bits and physically transmitted.

Note: Each layer adds specific information to help the data reach its destination correctly. These steps are reversed when the data arrives at the receiving device.

Practical Example:

Sending an Email

Let us walk through an example of Person A sending an email to Person B.

Step 1 (Application Layer): Person A interacts with an email application such as Gmail or Outlook and composes the message.

Step 2 (Presentation Layer): The email application prepares the data for transmission by encrypting it and formatting it appropriately.

Step 3 (Session Layer): A connection is established between the sender and receiver over the internet.

Step 4 (Transport Layer): The email data is broken into smaller segments. Sequence numbers and error-checking information are added to ensure reliability.

Step 5 (Network Layer): Packets are addressed and the best route for transfer is determined.

Step 6 (Data Link Layer): Data packets are encapsulated into frames. MAC addresses are added for local devices, and error detection is performed.

Step 7 (Physical Layer): Frames are transmitted as electrical or optical signals over a physical network medium, such as Ethernet cable or Wi-Fi.

Once the email reaches Person B (the receiver), the entire process is reversed. The data is decrypted, reassembled, and finally displayed in Person B's email client.

- **Map real-world examples to each layer (e.g. HTTP → Application, Ethernet → Data Link).**

OSI Model Real-World Mapping

The OSI model offers a seven-layer structure for understanding networking. It connects practical technologies from user-facing applications down to physical hardware into a coherent framework.

Below are real-world examples assigned to each OSI layer.

Layer 7: Application Layer (User Interface)

What it does: Provides the interface through which users interact with network services.

Real-world examples:

HTTP/HTTPS (web browsing), SMTP/IMAP (email), FTP (file transfers), SSH (remote login), DNS (domain name resolution), WhatsApp Web.

Scenario:

Opening a browser and typing `google.com` — this action uses HTTP/HTTPS.

Layer 6: Presentation Layer (Data Formatting)

What it does: Handles data translation, encryption, and compression so information can be understood by the receiving application.

Real-world examples:

SSL/TLS (encryption and security), JPEG/PNG (image formats), MP4/AAC (video and audio), JSON/XML (data structuring), ASCII (character encoding)

Scenario:

When you visit a banking website, your browser automatically decrypts the incoming data using TLS.

Layer 5: Session Layer (Session Management)

What it does: Establishes, maintains, and terminates communication sessions between two devices.

Real-world examples:

RDP (Remote Desktop Protocol), SQL queries, NetBIOS, web session IDs (used to keep users logged in).

Scenario:

Joining a Zoom meeting and keeping the audio/video stream active throughout the call.

Layer 4: Transport Layer (End-to-End Delivery)

What it does: Ensures complete data transfer with appropriate reliability and speed.

Real-world examples:

TCP (reliable delivery for web traffic and email), UDP (fast delivery for streaming, gaming, and VoIP).

Scenario:

WhatsApp voice calling uses UDP — speed is prioritized over perfect packet delivery.

Layer 3: Network Layer (Routing)

What it does: Determines the best path to send data from source to destination across different networks.

Real-world examples:

IP (IPv4 and IPv6), ICMP (used by ping), OSPF, BGP (Border Gateway Protocol).

Scenario:

Your home router uses IP addresses to route a Netflix data packet to your television.

Layer 2: Data Link Layer (Physical Addressing)

What it does: Transfers data between nodes on the same local network using hardware addresses.

Real-world examples:

Ethernet (802.3), Wi-Fi (802.11), MAC addresses, network switches, ARP (Address Resolution Protocol).

Scenario:

An Ethernet switch uses MAC addresses to forward data only to the intended device on your local network.

Layer 1: Physical Layer (Hardware Transmission)

What it does: Transmits raw bits (0s and 1s) over a physical medium.

Real-world examples:

Ethernet cables (Cat5e, Cat6), fiber optic cables, Wi-Fi radio waves, Bluetooth, USB connectors, hubs, network interface cards (NICs).

Scenario:

A Wi-Fi router sends radio signals through the air to your laptop.

Summary Table: Web Request Example (`https: //google.com)

OSI Layer	Component	Real-World Activity
7. Application	HTTP/HTTPS	Browser requests the webpage
6. Presentation	TLS/SSL	Data is encrypted before transmission
5. Session	Session ID	Connection state is maintained during the request
4. Transport	TCP	Reliable delivery handshake occurs (SYN/ACK)
3. Network	IP	Packets are routed using IP addresses
2. Data Link	Ethernet/Wi-Fi	Data is framed and MAC addresses are added
1. Physical	Cables/Radio	Bits are transmitted as electrical or light signals

➢ TASK: Functions of each OSI layer.

Here are the core functions of each OSI layer, presented in a clear, original, and professional manner.

Functions of Each OSI Layer (Layers 1–7)

Layer 1: Physical Layer

Primary role: Transmission of raw bits over a physical medium.

Key functions:

- Bit encoding and signaling (converting 0s and 1s into electrical, optical, or radio signals).
- Defining physical characteristics of cables, connectors, and interfaces.
- Managing data rates (transmission speed).
- Synchronizing sender and receiver clocks.
- Specifying transmission direction (simplex, half-duplex, full-duplex).

Layer 2: Data Link Layer

Primary role: Reliable node-to-node data transfer across a single network link.

Key functions:

- **Framing:** Grouping bits into frames with defined start and end boundaries.
- **Physical addressing:** Adding source and destination MAC addresses to frames.
- **Error detection and correction:** Identifying and retransmitting corrupted or lost frames.
- **Flow control:** Preventing a fast sender from overwhelming a slow receiver.
- **Media access control:** Determining which device can transmit on a shared medium.

Layer 3: Network Layer

- **Primary role**: End-to-end delivery of packets across multiple networks.

- **Key functions:**
- **Routing:** Selecting the best path from source to destination
- **Logical addressing:** Assigning unique IP addresses to identify devices across different networks
- **Packet forwarding:** Moving packets from one router to the next toward the destination
- **Fragmentation and reassembly:** Breaking packets into smaller units when needed
- **Error handling and diagnostics:** Using protocols like ICMP for network troubleshooting

Layer 4: Transport Layer

Primary role: Reliable, end-to-end communication between processes on different hosts.

Key functions:

- **Segmentation and reassembly:** Dividing application data into smaller segments and reassembling them at the destination.
- **Port addressing (service point addressing):** Using port numbers to deliver data to the correct application.
- **Connection management:** Establishing, maintaining, and terminating connections (TCP only).
- **Reliability control:** Acknowledging received data and retransmitting lost segments.
- **Flow control (end-to-end):** Preventing buffer overflow at the receiver side.

Layer 5: Session Layer

Primary role: Managing communication sessions between applications

Key functions:

- **Session establishment, maintenance, and termination:** Setting up and closing active communication sessions.
- **Synchronization:** Inserting checkpoints (sync points) into data streams to allow recovery after interruptions.
- **Dialog control:** Determining whose turn it is to transmit (half-duplex or full-duplex).
- **Session restoration:** Recovering interrupted sessions without restarting from scratch.
- **Activity management:** Grouping related exchanges into logical activities.

Layer 6: Presentation Layer

Primary role: Translating data between application format and network format.

Key functions:

- **Data translation:** Converting between different character encodings (e.g., ASCII, EBCDIC).
- **Encryption and decryption:** Securing data for transmission and decrypting upon arrival.
- **Compression and decompression:** Reducing the number of bits transmitted to save bandwidth.
- **Data formatting:** Structuring data using standards like JSON, XML, or binary formats.
- **Protocol conversion:** Translating between different application-layer protocols when needed.

Layer 7: Application Layer

Primary role: Providing network services directly to end-user applications.

Key functions:

- **User interface:** Allowing applications to request network services.
- **Resource sharing and access:** Enabling remote file, printer, and database access.
- **Email services:** Supporting message composition, transfer, and storage.
- **Directory services:** Providing name-to-address resolution (e.g., DNS).
- **Remote login:** Allowing users to access remote systems securely (e.g., SSH, Telnet).
- **File transfer:** Moving files between systems (e.g., FTP, SFTP).
- **Web services:** Delivering web pages and APIs (e.g., HTTP/HTTPS)

➢ TASK: Data flow process (encapsulation & decapsulation).

Data Encapsulation

Data Encapsulation is the process in which some extra information is added to the data item to add some features to it. We use either the OSI or the TCP/IP model in our network, and the data transmission takes place through various layers in these models. Data encapsulation adds the protocol information to the data so that data transmission can take place in a proper way. This information can either be added in the header or the footer of the data. The data is encapsulated on the sender's side, starting from the application layer to the physical layer. Each layer takes the encapsulated data from the previous layer and adds some more information to encapsulate it and some more functionalities with the data. These functionalities may include proper data sequencing, error detection and control, flow control, congestion control, routing information, etc.

Data De-encapsulation

Data De-encapsulation is the reverse process of data encapsulation. The encapsulated information is removed from the received data to obtain the original data. This process takes place at the receiver's end. The data is de-encapsulated at the same layer at the receiver's end to the encapsulated layer at the sender's end. The added header and trailer information are removed from the data in this process. The below diagram shows how header and footer are added and removed from the data in the process of encapsulation and de-encapsulation respectively. The data is encapsulated in every layer at the sender's side and also de-encapsulated in the same layer at the receiver's end of the OSI or TCP/IP model. Actually, we use different terms for the encapsulated form of the data that is described in the below mentioned diagram. Now, we will learn the whole process of encapsulation and de-encapsulation in the OSI and TCP/IP model step-by-step as mentioned in the below picture.

Encapsulation Process (At sender's side)

1. Step 1: The Application, Presentation, and Session layer in the OSI model, or the Application layer in the TCP/IP model takes the user data in the form of data streams, encapsulates it and forwards the data to the Transport layer. It does not necessarily add any header or footer to the data. But it is application-specific and can add the header if needed

2. Step 2: The Transport layer (in the OSI or TCP/IP model) takes the data stream from the upper layers, and divide it into multiple pieces. The Transport layer encapsulates the data by adding the appropriate header to each piece. These data pieces are now called as data segments. The header contains the sequencing information so that the data segments can be reassembled at the receiver's end.

3. Step 3: The Network layer (in the OSI model) or the Internet layer (in the TCP/IP model) takes the data segments from the Transport layer and encapsulate it by adding an additional header to the data segment. This data header contains all the routing information for the proper delivery of the data. Here, the encapsulated data is termed as a data packet or datagram.

4. Step 4: The Data-Link layer (in the OSI or TCP/IP model) takes the data packet or datagram from the Network layer and encapsulate it by adding an additional header and footer to the data packet or datagram. The header contains all the switching information for the proper delivery of the data to the appropriate hardware components, and the trailer contains all the information related to error detection and control. Here, the encapsulated data is termed as a data frame.

5. Step 5: The Physical layer (in the OSI or TCP/IP model) takes the data frames from the Data-Link layer and encapsulate it by converting it to appropriate data signals or bits (corresponding to the physical medium).

De-Encapsulation Process (At receiver's side)

1. Step 1: The Physical layer (in the OSI or TCP/IP model) takes the encapsulated data signals or bits from the sender, and de-encapsulate it in the form of a data frame to be forwarded to the upper layer, i.e., the Data-Link layer.

2. Step 2: The Data-Link layer (in the OSI or TCP/IP model) takes the data frames from the Physical layer. It de-encapsulates the data frames and checks the frame header whether the data frame is switched to the correct hardware or not. If the frame is switched to the incorrect destination, it is discarded, else it checks the trailer information. If there is any error in the data, data retransmission is requested, else it is de-encapsulated and the data packet is forwarded to the upper layer. \

3. Step 3: The Network layer (in the OSI model) or the Internet layer (in the TCP/IP model) takes the data packet or datagram from the Data-Link layer. It de encapsulates the data packets and checks the packet header whether the packet is routed to the correct destination or not. If the packet is routed to the incorrect destination, the packet is discarded, else it is de-encapsulated and the data segment is forwarded to the upper layer.

4. Step 4: The Transport layer (in the OSI or TCP/IP model) takes the data segments from the network layer and de-encapsulate it. It first checks the segment header and then reassembles the data segments to form data streams, and these data streams are then forwarded to the upper layers.

5. Step 5: The Application, Presentation, and Session layer in the OSI model, or the Application layer in the TCP/IP model takes encapsulated data from the Transport layer, de-encapsulate it, and the application-specific data is forwarded to the applications. This is all about the data encapsulation and de-encapsulation in computer.

Data Movement in the OSI Model: Adding and Removing Headers

When information travels between two devices over a network, it must pass through all seven OSI layers. This journey consists of two opposite but related processes:

- **Encapsulation (transmitter side):** Each layer wraps the incoming data with its own control information (mostly headers, sometimes trailers) before passing it downward.
- **Decapsulation (receiver side):** Each layer strips off its corresponding control information as data moves upward toward the user application.

Encapsulation – Building the Data Package for Transmission

Encapsulation starts when an application generates data. As that data descends through the layers, each layer adds a specific piece of control information.

Detailed Encapsulation Steps

Phase	Layer	What Happens	Resulting Data Unit
1	Application (7)	the user's application creates the original payload	Data
2	Presentation (6)	the data may be encrypted, compressed, or reformatted	Data
3	Session (5)	Session identifiers and synchronization markers are inserted	Data
4	Transport (4)	the data stream is split into smaller pieces attached to each piece	Segment
5	Network (3)	a network header is prefixed to each segment	Packet.
6	Data Link (2)	A data link header is added to the front, and a trailer is attached to the end	Frame
7	Physical (1)	the complete frame is turned into a stream of bits and sent across the medium	Bits

Visual of Encapsulation

Sender – Top to Bottom

Original application data

[Transport Header | Original Data] ← This is a segment

[Network Header | Transport Header | Original Data] ← This is a packet

[Data Link Header | Network Header | Transport Header | Original Data | Data Link Trailer] ← this is a frame

[1 0 1 1 0 0 1 0 1 1 0 0 ...] ← Bits on the wire

Part 2: Decapsulation – Unwrapping the Data at the Destination

Decapsulation begins when the receiving device picks up the bits from the physical medium. As the data climbs upward, each layer removes the header (and trailer) that its counterpart added on the sender side.

Detailed Decapsulation Steps

Physical (1):

The device reads incoming bits and groups them into a frame; the frame is then handed to the Data Link Layer.

Data Link (2): The layer examines and removes the data link header and trailer; it verifies the CRC for errors; the remaining packet is passed upward to the Network Layer.

Network (3): The network header is read and stripped away; the destination IP address is checked; the extracted segment is sent to the Transport Layer

Transport (4): The transport header is processed and removed; segments are reassembled into the correct order; missing or damaged data is identified; the recovered data is passed to the Session Layer.

Session (5): Session control fields are read and removed; synchronization points are handled; the data continues upward to the Presentation Layer.

Presentation (6): The data is decrypted, decompressed, or converted back to its original format as needed; clean data is delivered to the Application Layer.

Application (7): The original message is finally presented to the user application.

Visual of Decapsulation

Receiver – Bottom to Top

Incoming bits from the network medium

[Data Link Header | Network Header | Transport Header | Original Data | Data Link Trailer] ← Frame

(Data Link Layer removes header + trailer)

[Network Header | Transport Header | Original Data] ← Packet

(Network Layer removes header)

[Transport Header | Original Data] ← Segment

(Transport Layer removes header)

Original application data

User application receives the data

Definition and Core Concepts

Encapsulation is the process where a sender adds headers and trailers to data units as they move down the Open Systems Interconnection (OSI) or Transmission Control Protocol/Internet Protocol (TCP/IP) model layers. Each layer creates a new, larger Protocol Data Unit (PDU) by wrapping the previous layer's data with its own control information. Decapsulation performs the inverse operation. The receiver removes headers and trailers from data units as they move up the network stack layers. This reveals the original data payload for processing by the next higher layer.

OSI and TCP/IP Models

The OSI model provides a seven-layer conceptual framework for network communication. The TCP/IP model uses four layers that map to specific OSI layers. Both models define how encapsulation and decapsulation occur at each protocol level.

Protocol Headers and Trailers

Headers contain control information specific to each layer's protocol. This includes addressing information, sequence numbers, flags, and error detection codes. Trailers appear primarily at the Data Link Layer for error checking through mechanisms like Frame Check Sequence (FCS).

Protocol Data Units

Each network layer uses specific terminology for its data unit:

- Application Layer: Data or Message
- Transport Layer: Segment (TCP) or Datagram (UDP)
- Network Layer: Packet
- Data Link Layer: Frame
- Physical Layer: Bits

How Encapsulation Works

Encapsulation begins at the Application Layer when a user application generates data for transmission. The data moves down through each layer of the network stack, with each layer adding its specific header information.

Application Layer Processing

The application creates raw data such as HTTP requests, email messages, or file transfers. This data contains no network addressing or transport information.

Transport Layer Addition

The Transport Layer adds either a TCP or UDP header. TCP headers include source and destination port numbers, sequence numbers, acknowledgment numbers, window size, and control flags. UDP headers contain only source port, destination port, length, and checksum fields.

Network Layer Wrapping

The Network Layer encapsulates the transport segment within an IP header. This header contains source and destination IP addresses, Time to Live (TTL) values, protocol identifiers, and fragmentation information. The resulting PDU becomes an IP packet.

Data Link Layer Framing

The Data Link Layer creates a frame by adding both a header and trailer. The header includes source and destination Media Access Control (MAC) addresses and frame type information. The trailer contains the Frame Check Sequence for error detection.

Physical Layer Transmission

The Physical Layer converts the complete frame into electrical signals, light pulses, or radio waves for transmission across the physical medium.

How Decapsulation Works

Decapsulation reverses the encapsulation process when data arrives at the destination device. The receiving network stack processes the data from the Physical Layer upward through each protocol layer.

Physical Layer Reception

The Physical Layer receives the transmitted signals and converts them back into digital frame data for processing by higher layers.

Data Link Layer Processing

The Data Link Layer examines the frame header to verify the destination MAC address matches the receiving interface. It performs error checking using the FCS trailer and removes both the header and trailer before passing the packet to the Network Layer.

Network Layer Examination

The Network Layer processes the IP header to verify the destination IP address. It checks TTL values, handles fragmentation if necessary, and removes the IP header before passing the segment to the Transport Layer.

Transport Layer Handling

The Transport Layer examines port numbers to determine the destination application. For TCP connections, it handles sequence numbers, acknowledgments, and flow control. The layer removes the transport header and delivers the data to the appropriate application.

Application Layer Delivery

The Application Layer receives the original data in its unmodified form for processing by the end-user application.

Key Features and Components

Several critical components enable encapsulation and decapsulation processes to function reliably across diverse network environments.

Sender and Receiver Endpoints

Network devices act as senders when transmitting data and receivers when processing incoming transmissions. Each device must implement complete network stacks to handle both encapsulation and decapsulation operations.

Network Stack Implementation

Operating systems implement network stacks as software modules that correspond to OSI or TCP/IP model layers. These modules interact through defined interfaces to process data units appropriately.

Header Information Fields

Headers contain essential control information:

- **Addressing**: Source and destination identifiers at each layer.
- **Sequencing**: Order information for reliable delivery.
- **Control Flags**: Protocol-specific operational parameters.
- **Error Detection**: Checksums and validation codes.

Trailer Components

Trailers primarily appear at the Data Link Layer and contain Frame Check Sequence values calculated from the entire frame contents. Some protocols use trailers for additional error detection or security information.

Use Cases and Applications

Encapsulation and decapsulation enable all forms of network communication through their fundamental role in data transmission protocols.

Standard TCP/IP Communication

Web browsing, email, file transfers, and remote access all rely on these processes. Each application generates data that undergoes complete encapsulation for transmission and decapsulation upon receipt.